"Is Stephen Curry really a guard? A new perspective on player typologies using functional data analysis": Reviewer Responses

General/ Overall Responses

We thank the associate editor and the two reviewers for their thorough and detailed review of our manuscript. We have addressed each review point by point, making appropriate changes in the revised manuscript. As a result, we believe that the revised manuscript is greatly improved. Responses to each of the reviewer's points are contained in the remainder of this document. We look forward to your final decision with anticipation.

Associate Editor

Comment # 0.1

The author has effectively addressed some of the comments from the initial submission. Two new referees have reviewed the paper and give several suggestions for further revision. In addition, the authors should consider the following points.

Reply:

We thank the associate editor for the careful consideration of our manuscript. We have addressed their points and those of the two reviewers and improved the manuscript as a result. Our responses are presented in a point-by-point fashion below.

Comment # 0.2

The MFPCA approach was described more clearly in the revision, which leads to an important point about visualization. In equation (2), the eigenfunctions $\phi_k(s,t) \in \mathbb{R}^2$ are bivariate, with one component corresponding to made and the other to missed shots. Doing MFPCA instead of marginal FPCA is nice since variability in made and missed shot locations are taken together, yielding a single set of scores for each player to cluster. However, there is no indication in the figures regarding how much each eigenfunction loads onto made and missed shots. In other words, it is possible that $\phi_1(s,t)$ loads mostly onto made shots and almost none onto missed shots, or vice versa, but there is no way to tell this from the current analysis because the normalization (a linear transformation, I assume) is applied separately to every panel in each figure. It would be more informative to apply this transformation to the combined data across made and missed shots - that is, normalization is applied to each column. Also, it would be helpful to report how much each component contributes to the variability explained in made/missed shots separately, in addition to the joint figures currently reported.

Reply:

We agree with this comment. The separate normalization of each panel is useful for visualizing spatial patterns, but hides the relative magnitude of the made-shot and missed-shot components. We have therefore change the visualization. The normalization is now applied jointly to the made-shot and missed-shot surfaces within each principal components. More precisely, for each column of the figures, we computed the maximum value of the densities and divided the made-shot and missed-shot densities by this value.We do not see much difference with the previous version, indicating that the load of each eigenfunction onto made and missed shots is balanced.

I am not sure to understand the last sentence of the comment. I believe we do not need to do anything except report the loading. Ramsay and Silverman (2005) do this very simply. If $\boldsymbol{\phi}(t) = (\phi^{(1)}(t), \phi^{(2)}(t))^2$ is a bivariate eigenfunction, then they worked out that an appropriate "relative contribution" in each dimension was:

$$\int_{\mathcal{T}} (\phi^{(1)}(t))^2 dt \quad \text{and} \quad \int_{\mathcal{T}} (\phi^{(2)}(t))^2 dt$$

which is already a relative measure since

$$\int_{\mathcal{T}} (\phi^{(1)}(t))^2 dt + \int_{\mathcal{T}} (\phi^{(2)}(t))^2 dt = 1$$

by the orthonormality property. I think all you need would be to display these numbers on the plot. **To Discuss – why do we need to normalize at all?**

COMMENT # 0.3

Although the author has argued that lack of consensus in a suitable spatial grid leads to subjectivity in previous analyses, there is no apparent justification for the choice of a 201×201, equally spaced grid for the MFPCA. In addition, the density estimation induces further subjectivity by choice of a bandwidth. Is there a way to assess sensitivity of the results and interpretations to these choices? It seems the choice is no less arbitrary for density representations compared to previous methods.

Reply:

The 201×201 grid is a device used to evaluate smooth density functions and approximate eigenfunctions. Thus, the grid is not used to define regions, impose boundaries or create discrete shooting categories. The considered data are smooth functional densities over the offensive half-court. The chosen grid is sufficiently fine such that any discretization error is negligible in the context of our statistical analysis.

I would reduce/ remove below, and yes do a sensitivity to Grid size but just in calculating the PCs. The grid is the resolution at which those smooth functions are representative computationally. The arbitrariness, mentioned in the manuscript, is numerical and not conceptual. At this resolution, the grid is sufficiently dense to represent the smoothed density surfaces while remaining computationally efficient. Increasing the grid resolution would mainly refine the numerical approximation of the same continuous objects.

Regarding the bandwidth, Silverman's rule is a standard, well-established and regularly-used bandwidth selection method. It is theoretically grounded and avoids the computational expense of re-smoothing a number of times as is required in resampling approaches (e.g., cross-validation)

Any idea of what to do, except cross-validation? I would do a MINIMAL sensitivity – maybe a couple of observations via CV, and just trial one larger grid for FPCA and show results are near identical

In our approach, the grid is merely an implementation tool, and the bandwidth is chosen by a standard automatic rule.

Should we show different grids? Or different bandwidth? It will be quite long to do so.

COMMENT # 0.4

Related to the previous point, it is clear that densities are able to examine patterns without the influence of volume. However, how would removing volume information prove valuable in making performance-related decisions in terms of coaching, scouting, or training, for example? Are there other solutions that could be more effective, such as stratifying by volume rather than removing it altogether? Please address this more completely in the discussion.

Reply:

Ed To Do: Double check and either reference discussion or add part. Our density-based representation is intended to characterize where a player tends to shoot, conditional on the player having taken shots, rather than how often the player shot shoots overall. This is useful in making performance-related decisions when the goal is to compare spatial shooting tendencies independently of playing time, number of games or usage. For example, two players with very different total shot counts can still have similar spatial tendencies. Our approach allows this to be detected.

We should add something in the discussion.

We could add intensity surfaces as a third coordinate in the vector-valued stochastic process. I am not sure it is a good idea to stratify by volume, because the estimation of the densities depend on the number of shots we have for the player. So, players with less shots will have their densities less accurately estimated. Also, how do we choose the strate? Alternatively, shot volume could also be included as an additional feature in the clustering step.

COMMENT # 0.5

The choice of $k = 5$ is clearly not data-driven. What would a true data-driven clustering look like? Certainly there are more than 5 shot pattern styles, so what information can we really gather from the clustering beyond observing that they clusters with $k = 5$ are poorly aligned with player position? Also, although several metrics on the clustering are reported (ARI, Silhouette score), the relevance of the values is not discussed or their bearing on the *reliability and impact of the clustering results.*

Reply:

Why should everything be data-driven? I think it makes sense to have $k = 5$... I agree..

The choice of $k = 5$ is motivated by the comparison with the five NBA defined positions, not by a data-driven cluster-selection criterion. Generally, it is accepted that the five players on the NBA court occupy five distinct positions. We use this knowledge to constrain our problem which facilitates a comparison of our five-class, data-driven taxonomy with the conventional five-class NBA typology. While our methodology itself is empirical and data driven, the five-cluster constraint is informed by sport-specific knowledge, which we see as a benefit in practical data analysis settings. That being said, we now provide a sensitivity analysis to the number of clusters in the revised supplement...

The clustering should be interpreted as an illustrative topology of players and not as evidence that there are exactly five natural shooting styles in the NBA.

Here, we'll probably have to add some silhouette plot and return the number of clusters that maximise the silhouette. I think so, just to appease reviewer!

Reviewer # 1

Comment # 1.1

The current research is characterizing NBA player shooting behavior using functional data analysis. The authors treat made and missed shot locations as bivariate functional data on the offensive half-court and perform a multivariate functional principal component analysis to extract dominant shooting patterns. Then, using k-medoids clustering on the resulting player-specific scores, they generate a data-driven typology of player shooting styles, questioning the validity of traditional NBA positional labels. Several conceptual and technical issues limit both the novelty and the inferential credibility of the results. I therefore recommend a revision before the paper can be considered for publication.

Reply:

Thank you for the detailed review of the manuscript. We have addressed each of the concerns point by point below.

Comment # 1.2

While the paper is intended for a quantitative audience, some statistical steps, especially the MFPCA, are described at a relatively informal level. There is no discussion of convergence or uncertainty in the functional principal components.

Reply:

Given the applied nature of the paper, we have not presented convergence results for MFPCA. However, a full theoretical treatment of the method can be found in Happ and Greven, 2018. While we are primarily concerned with describing the variation in an entire population of NBA players, rather than making inferences using a sample, uncertainty quantification for the MFPCs is not of primary concern either. However, to assess stability/ sensitivity of our approach, we added a simple bootstrap step (see comment 0.6).

Comment # 1.3

The authors do not evaluate the stability of their clustering. Furthermore, they do not provide external validation (e.g., prediction, test–train splits, comparison with existing clustering approaches). Please consider using metrics such as Adjusted Rand Index or Silhouette scores *to assess cluster quality. Alternatively, compare results with simpler clustering methods (e.g., k-means on shooting frequency by location bins) to demonstrate added value.*

Reply:

Adjusted Rand Index (ARI) and Silhouette coefficients have been computed. ARI between the NBA clusters and the proposed clusters are close to 0 but this is expected as we are looking for another typology than usual NBA players. The Silhouette coefficient for the k-medoids with equal weights is equal to 0.04, while the Silhouette coefficient for the k-medoids with variance weights is equal to 0.4. We also computed the Silhouette coefficient of the NBA clustering using the distance matrix used for the k-medoids with equal weights (-0.05) and using the distance matrix used for the k-medoids with variance weights (-0.1).

COMMENT # 1.4

While contrasting the clusters with NBA-defined positions is informative, the resulting disagreement is not deeply analyzed. Are there specific cases where agreement is worse?

Reply:

We do highlight that there is agreement for the center category:

> *"The only exception is the NBA-defined 'center' category, which aligns with the cluster 1 in the k-medoids results when weighted by the variance explained."*

Since the majority of both clusters and positions (each axis of the confusion matrix) disagree, it is difficult to pinpoint the most disagreement.

COMMENT # 1.5

The paper uses a 2D KDE with Gaussian kernel and Silverman's rule but does not specify whether the bandwidth is optimized globally, per player, or per component.

Reply:

A different bandwidth is considered for each component (missed and made shots chart) of each player. We have added a sentence to the main text to clarify this.

COMMENT # 1.6

The model jointly treats made and missed shots, but no attention is given to efficiency metrics (i.e., shooting percentage), which are arguably central to evaluating shot quality.

Reply:

We have now added the following to the discussion to acknowledge this point:

> *"Secondly, this work focused on characterising the spatial patterns of made or missed shots. The normalization in the density estimation step, performed separately for the made and missed shots, inherently removes information about total shot volume or shooting efficiency (e.g., percentage chance of success). Future work could include an overall measure of shot volume or shooting efficiency to improve the characterization of players' behaviours."*

Reviewer # 2

Comment # 2.1

Alternative clustering approaches I think the authors should include the discussion of alternative clustering methods in the main text (or in the appendix).

Reply:

We thank the reviewer for the suggestion. In the revised manuscript, we have added a discussion of alternative clustering approaches. We also clarified why the k-medoids algorithm was selected for the main analysis. Specifically, the k-medoids algorithm has the advantage that each cluster center is an actual player rather than an artificial average point. This makes the resulting clusters easier to interpret in this context, because each cluster can be summarized by a representative player.

We have also clarified that the choice of the method is not intrinsic to the proposed framework. The representation provides a general low-dimensional summary of each player's shooting pattern and different clustering algorithms (or other statistical methods) could be applied to the representation depending on the goal of the analysis. For example, hierarchical clustering is useful when we want to study nested similarities between players; mixture models are useful when a probabilistic assignment is desired and a density-based method is useful to identify outliers.

We have added the following sentences to the Discussion section:

> Although, we use the k-medoids clustering algorithm in our analysis, the MFPCA representation is not tied to a particular clustering algorithm. Actually, once each player's shot chart has been summarized by a finite vector of scores, all of the standard clustering algorithms could be applied. For example, k-means clustering could be used to identify groups around average score profiles, hierarchical clustering could show nested similarities among players, mixture models could provide probabilistic cluster memberships and density-based methods could be use in case of outliers. We use the k-medoids algorithm because the center of each cluster is an actual player from the NBA, rather than an artificial mean profile. This provides a direct basketball interpretation of each cluster through its medoid.

Comment # 2.2

Since the data span multiple seasons (2018-19 through 2023-24), shot selection can change across seasons (and also within a season), due to numerous reasons (e.g., changes in team, *coaching, usage rate, playing style, etc.). This is perhaps outside the scope of the paper, but I'm interested in seeing a discussion on whether your method can be extended to handle this. I think having a couple of paragraphs on this in the Discussion would make the paper more intriguing to interested readers. For example, young players may enter the league without a reliable 3-point shot and therefore prefer scoring near the basket. Over time, they may develop a consistent 3-point shot and become more comfortable taking 3s. A similar pattern may apply to centers. Players such as Al Horford and Brook Lopez once favored post-ups and shots around the rim but later expanded their range and became good at 3-point shooting. (These are just examples of the top of my head; these developments by Horford and Lopez occurred before the sample of seasons considered in this paper.) Alternatively, have you also considered clustering at the player-season level instead of just one single aggregated shot chart for each player? Extending this to a player-season level could further validate your finding of empirical clusters having low agreement with official positions. For instance, it could help quantify the role/positional evolution of a player (related to what I described above).*

Reply:

We thank the reviewer for this suggestion. We agree that aggregating shots across multiple seasons, while providing a stable summary, may hide potentially meaningful temporal changes. A similar procedure could be applied after redefining each observational unit from "player" to "player-season". The resulting MFPCA scores could then be analyzed longitudinally, considering that the observations are not independant. We could then model score trajectories as a function of age or something else. Clustering the player-season scores would also allow players to move between shooting-style groups over time and quantifying role evolution. While being outside the scope of the paper, we added a paragraph in the Discussion section explaining how the proposed method could be extended to account for season-to-season variability.

We have added the following sentences to the Discussion section:

> We aggregated shooting behavior across the 2018-2019 through 2023-2024 regular season. However, this aggregation may mask meaningful season-to-season changes in shooting behavior, as a player's shot selection can evolve with time due to development, injuries, coaching strategy, etc. A natural extension of our framework would be to perform the analysis at the player-season level rather than at the aggregated player level. In this setting, the MFPCA scores could be used to represents the evolution of shooting-style over time. This could help to quantify the temporal evolution of a player's role, identify players whose shooting patterns change across seasons, and examine whether these changes correspond to team

changes, coaching changes or something else. This could also strengthen the comparison between empirical shooting-pattern clusters and conventional NBA positions. We could assign a cluster to each player-season and study the stability of these assignments.

Great reply but just TBC on last point... think we have it wrong way around (allocate player-season to cluster rather than cluster to player-seasondots)

References

Happ, C., & Greven, S. (2018). Multivariate Functional Principal Component Analysis for Data Observed on Different (Dimensional) Domains. *Journal of the American Statistical Association*, *113*(522), 649–659.